Day - 12
Python-While Loop

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What is a While Loop?

- A **while loop** repeats a task **as long as a condition is true**.
- Example:
  "I will keep drinking water **while** I'm thirsty."
  If the condition becomes false, the loop stops.

- Technical Definition
  - A while loop is a control structure that executes a block of code **repeatedly** as long as the given condition remains **True**.

  Syntax:

  ```
  while condition:
    # repeated code
  ```

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Why While Loop is Important?

- We **do not know** how many times the loop should run
- Condition-based repetition
- Reading data until file ends
- Cleaning data until condition is met
- Taking user input continuously

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Basic Example

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

Infinite Loop

```
x = 1
while x < 5:
    print("Hello")  # x never changes → infinite loop!
```

Example

Example – Countdown

```
n = 5
while n > 0:
    print(n)
    n -= 1
```

Loop Until User Stops :

```
name = ""
while name == "":
    name = input("Enter your name: ")

print("Welcome,", name)
```

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Example – Ask until valid number

```
num = ""

while not num.isnumeric():
    num = input("Enter a number: ")

print("Valid number:", num)
```

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Loop Through List Using While

```
items = ["Laptop", "Mouse", "Keyboard"]
i = 0

while i < len(items):
    print(items[i])
    i += 1
```

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While Loop with Break

```
num = 1

while num <= 10:
    if num == 5:
        break
    print(num)
    num += 1
```

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While Loop with Continue

```
x = 0
while x < 10:
    x += 1
    if x % 2 == 0:
        continue
    print(x)
```

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Password Retry System

```
password = ""
attempts = 0

while password != "admin123" and attempts < 3:
    password = input("Enter password: ")
    attempts += 1

if password == "admin123":
    print("Login Successful")
else:
    print("Account Locked")
```

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Assignments

✓ **Basic**

Print numbers from 10 to 1

Ask user their favorite color until they enter something

✓ **Intermediate**

Count vowels in a string using while

Loop through list using while

✓ **Advanced**

Build a login system with 3 attempts

Ask user to enter prices until they type “STOP”, then calculate total